Problem 8 – Humanize

1 The statement and the plan

Let M be a compact connected smooth (C^∞) manifold, let X be a C^∞ (real) vector field on M, and let $g_t : M \to M$, $t \in \mathbb{R}$, be the flow it generates (globally defined since M is compact). Write $\mathcal{D}(M) = C^\infty(M)$ for the smooth complex valued test functions on M and $\mathcal{D}'(M)$ for the distributions; the Lie derivative $\mathcal{L}_X$ acts on distributions by duality.

Definition 1.1. X is *globally hypoelliptic* (GH) if every distribution $u \in \mathcal{D}'(M)$ such that $\mathcal{L}_X u \in C^\infty(M)$ is itself in $C^\infty(M)$.

Main Theorem. If X is globally hypoelliptic, then the flow g_t has zero topological entropy.

The argument is organized as follows; the only genuinely external input is the Greenfield–Wallach structure theorem, quoted precisely as Theorem 2.1 below and used only for its topological consequence ("the flow extends to a smooth action of a compact torus").

(1) **The structure theorem** (Section 2): global hypoellipticity forces the flow g_t to be a one-parameter subgroup of a smooth action of a compact torus $\mathbb{T}^k$ on M (Theorem 2.1, Greenfield–Wallach). It follows (Lemma 2.3) that after averaging an arbitrary Riemannian metric over the compact group, the maps g_t are all isometries, hence the family $\{g_t\}_{t\in\mathbb{R}}$ is equicontinuous.

(2) **Equicontinuous flows have zero entropy** (Section 3): recall Bowen's definition of topological entropy (Definition 3.1); then (Lemma 3.2) an equicontinuous map on a compact metric space has zero entropy, since (n, ε)-separated sets have cardinality bounded independently of n.

We also give consistency checks (Remark 2.2) showing the statement is non-vacuous (Diophantine linear flows on $\mathbb{T}^d$ satisfy the hypothesis and have zero entropy) and that its hypotheses are sharp (vector fields with zeros, periodic orbits, or hyperbolic pieces never satisfy it).

2 From global hypoellipticity to a compact group action

The analytic hypothesis GH is extremely rigid. The precise structure it forces is the content of the following theorem of Greenfield and Wallach, which we state in the form we shall use.

Theorem 2.1 (Greenfield–Wallach)**.** *Let M be a compact connected smooth manifold and X a smooth vector field on M which is globally hypoelliptic (Definition 1.1), with flow g_t. Then there exist*

- *an integer $k \geq 1$, the torus $\mathbb{T}^k = \mathbb{R}^k/\mathbb{Z}^k$, and a smooth action $\mu : \mathbb{T}^k \times M \to M$;*

- *a real vector* $\omega = (\omega_1, \dots, \omega_k) \in \mathbb{R}^k$ *which is* Diophantine*, i.e. there are* $C, \tau > 0$ *such that*

$$\Big| \sum_{j=1}^{k} m_j \, \omega_j \Big| \;\geq\; C \, \|m\|^{-\tau} \qquad \textit{for all } 0 \neq m \in \mathbb{Z}^k;$$

such that, writing $\mu_a(x) = \mu(a, x)$ *for* $a \in \mathbb{T}^k$*,*

$$g_t \;=\; \mu_{\, t\omega \bmod \mathbb{Z}^k} \qquad \textit{for all } t \in \mathbb{R}.$$

Thus the flow of a globally hypoelliptic vector field is exactly a Diophantine one-parameter subgroup of a smooth compact torus action. (*Greenfield–Wallach,* Globally hypoelliptic vector fields*, Amer. J. Math.* ***94*** *(1972), 428–430. The two ingredients in the proof are: (i) the functional analysis of the equation* $\mathcal{L}_X u = f$*, due to Treves, which under GH produces a smooth nowhere-vanishing function of "period along orbits" and hence a smooth circle-valued first integral* F *with* $F(g_t x) = e^{it} F(x)$*; (ii) the observation that such an* F *exhibits* g_t *as the restriction of a* $\mathbb{T}^1$*-action to an orbit of a coisotropic structure, recursively a* $\mathbb{T}^k$*-action. For the present application only the* existence of the smooth compact torus action extending $\{g_t\}$ *is used.*)

Remark 2.2 (Consistency checks)**.** The statement is meaningful and non-trivial in both directions.

(a) *The zero field is not globally hypoelliptic*: $\mathcal{L}_0 u = 0 \in C^\infty$ for every $u \in \mathcal{D}'(M)$, but not every distribution is smooth. (Its flow is Id for all t, which also has zero entropy, so this borderline case would not affect the conclusion.)

(b) *Diophantine linear flows on* $\mathbb{T}^d$ *satisfy the hypothesis*: for $X = \sum_j \omega_j \partial_{x_j}$ on $\mathbb{T}^d$ with ω Diophantine, equation $\mathcal{L}_X u = f \in C^\infty$ gives, in Fourier coefficients, $\hat{u}(m) = \hat{f}(m) \big/ \big(2\pi i \sum_j m_j \omega_j\big)$ for $m \neq 0$. Since $\hat{f}(m)$ decays faster than any polynomial and the denominator is at worst polynomially small, $\hat{u}(m)$ decays faster than any polynomial, so u is smooth. Hence X is GH; and indeed its flow is an isometric translation of the flat torus, with zero entropy. Conversely, a *Liouville* ω produces divisors that are super-polynomially small along a sequence $m^{(\ell)}$, and one then constructs a smooth f whose formal solution is a genuine non-smooth distribution, so GH fails: this is exactly the dividing line in Theorem 2.1.

(c) *Vector fields with a zero, a periodic orbit, or a hyperbolic piece are never globally hypoelliptic*: they admit non-smooth invariant measures or distributions (e.g. Dirac masses and their transports) which lie in $\mathcal{L}_X^{-1} C^\infty \setminus C^\infty$. Consistently, their flows cannot extend to any compact group action, because orbits of compact group actions form an equicontinuous family (Lemma 2.3 below), which forbids saddles, attractors, or expansion.

The topological consequence of Theorem 2.1 that we need is the following elementary lemma.

Lemma 2.3. *Let* G *be a compact Lie group acting smoothly on the compact manifold* M *by* $\mu : G \times M \to M$*, and suppose the flow of* X *satisfies* $g_t = \mu_{a(t)}$ *for a continuous one-parameter subgroup* $a : \mathbb{R} \to G$*. Then there is a Riemannian metric on* M *such that every* g_t *(indeed every* μ_h*,* $h \in G$*) is an isometry. In particular, the family* $\{g_t\}_{t \in \mathbb{R}}$ *is equicontinuous for the induced distance* d*: for every* $\varepsilon > 0$*, taking* $\delta = \varepsilon$*,*

$$d(x, y) < \varepsilon \implies d\big(g_t x, g_t y\big) = d(x, y) < \varepsilon \qquad \textit{for all } x, y \in M,\ t \in \mathbb{R}.$$

Proof. Start with any smooth Riemannian metric γ_0 on M and average it over G with respect to the Haar probability measure dh:

$$\gamma_x(v,w) \coloneqq \int_G \gamma_0\big(D\mu_h(v), D\mu_h(w)\big)\, dh, \qquad v,w \in T_xM.$$

A convex combination of positive-definite inner products is positive definite, and smoothness of the action gives smoothness in x, so γ is a smooth Riemannian metric. It is G-invariant: for $k \in G$ and $v,w \in T_xM$, using right-invariance of Haar measure on the compact (hence unimodular) group,

$$(\mu_k^*\gamma)_x(v,w) = \int_G \gamma_0\big(D\mu_{hk}(v), D\mu_{hk}(w)\big)\, dh = \int_G \gamma_0\big(D\mu_h(v), D\mu_h(w)\big)\, dh = \gamma_x(v,w).$$

Hence every μ_k, in particular every $g_t = \mu_{a(t)}$, is an isometry of the associated distance d, so $d(g_tx, g_ty) = d(x,y)$ for all t, proving equicontinuity with $\delta = \varepsilon$. □

3 Equicontinuous flows have zero topological entropy

We use Bowen's separated-set definition and record the standard basic facts.

Definition 3.1 (topological entropy)**.** Let (M,d) be a compact metric space and $\varphi : M \to M$ continuous. For $n \in \mathbb{N}$ and $\varepsilon > 0$, a set $E \subset M$ is (n,ε)-*separated* if for every pair of distinct points $x,y \in E$ there is an integer j with $0 \le j < n$ and $d(\varphi^j x, \varphi^j y) \ge \varepsilon$. Let $s(n,\varepsilon)$ be the maximal cardinality of an (n,ε)-separated set, and set

$$h_{\mathrm{top}}(\varphi) \;=\; \lim_{\varepsilon\to 0} \limsup_{n\to\infty} \frac{1}{n} \log s(n,\varepsilon).$$

This value is independent of the choice of compatible metric d on the compact space M and equals the topological entropy defined via open covers (Bowen; see Walters, *An Introduction to Ergodic Theory*). The topological entropy of a flow g_t is defined as that of its time-one map, $h_{\mathrm{top}}(g) \coloneqq h_{\mathrm{top}}(g_1)$.

Lemma 3.2. *Let (M,d) be a compact metric space and $\varphi : M \to M$ a continuous map whose iterates $\{\varphi^j\}_{j\ge 0}$ form an equicontinuous family. Then $h_{\mathrm{top}}(\varphi) = 0$.*

Proof. Fix $\varepsilon > 0$. By equicontinuity of the iterates, there is a $\delta = \delta(\varepsilon) > 0$ such that

$$d(x,y) < \delta \implies d\big(\varphi^j x, \varphi^j y\big) < \varepsilon \quad \text{for every } j \ge 0.$$

Contrapositively, if $x \ne y$ belong to an (n,ε)-separated set then $d(x,y) \ge \delta$: the pair is separated at *time zero* already. Hence every (n,ε)-separated set is a δ-separated subset of the compact metric space (M,d), in the ordinary metric sense, regardless of n. Cover M by finitely many open balls of radius $\delta/2$; each such ball contains at most one point of any δ-separated set (two points in the same ball would be at distance $< \delta$), so the cardinality of any δ-separated set is at most the number $C(\delta)$ of balls in this cover. Therefore

$$s(n,\varepsilon) \;\le\; C\big(\delta(\varepsilon)\big) \;<\; \infty \qquad \text{for every } n \ge 1,$$

and, for this fixed ε,

$$\limsup_{n\to\infty} \frac{1}{n} \log s(n,\varepsilon) \;\le\; \limsup_{n\to\infty} \frac{1}{n} \log C\big(\delta(\varepsilon)\big) \;=\; 0.$$

Letting $\varepsilon \to 0$ gives $h_{\mathrm{top}}(\varphi) = 0$. □

Remark 3.3. If the flow consists of isometries, then Lemma 3.2 applies to $\varphi = g_t$ for every fixed t, giving $h_{\mathrm{top}}(g_t) = 0$ for all $t \in \mathbb{R}$. This is consistent with the standard identity $h_{\mathrm{top}}(g_t) = |t|\, h_{\mathrm{top}}(g_1)$ for flows, and shows the conclusion holds under every convention for the entropy of a flow.

4 Proof of the Main Theorem

Proof of the Main Theorem. Let X be a globally hypoelliptic smooth vector field on the compact connected manifold M, with flow g_t.

Step 1 (structure). By the Greenfield–Wallach theorem (Theorem 2.1), there is a smooth action μ of a compact torus $\mathbb{T}^k$ on M and a Diophantine vector $\omega \in \mathbb{R}^k$ such that $g_t = \mu_{t\omega}$ for all $t \in \mathbb{R}$: the flow is a one-parameter subgroup of the compact abelian group $G = \mathbb{T}^k$ acting smoothly on M.

Step 2 (equicontinuity). By Lemma 2.3, there exists a Riemannian metric γ on M with induced distance d such that every g_t is an isometry of (M, d). In particular the iterates $\{g_1^j\}_{j\geq 0} = \{g_j\}_{j\geq 0}$ of the time-one map form an equicontinuous family (with $\delta = \varepsilon$).

Step 3 (zero entropy). By Lemma 3.2 applied to $\varphi = g_1$ on the compact metric space (M, d), one has $h_{\mathrm{top}}(g_1) = 0$, hence by definition $h_{\mathrm{top}}(g) = 0$. By Remark 3.3, in fact $h_{\mathrm{top}}(g_t) = 0$ for every t. □

Remark 4.1. The hypothesis is essential: flows with zeros, periodic orbits, or any hyperbolic behavior have positive entropy very frequently and are never globally hypoelliptic (Remark 2.2(c)). The conclusion is non-vacuous: Diophantine linear flows on $\mathbb{T}^d$ (Remark 2.2(b)) are globally hypoelliptic, minimal and uniquely ergodic, and indeed have zero entropy, as they are isometries of the flat metric. Nothing beyond compactness and connectedness of M is used; the case $M = S^1$ is included.

References

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[2] F. Treves, *Topological Vector Spaces, Distributions and Kernels*, Academic Press, New York, 1967, §38.

[3] P. Walters, *An Introduction to Ergodic Theory*, Graduate Texts in Mathematics **79**, Springer-Verlag, New York, 1982.